

HDOC-4

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Question:

The following problem is to find the product of two integers without using the multiplication operator. Your teacher has given an incorrect algorithm and an incorrect flowchart for this problem. Study both carefully and complete the tasks given below.

Task 1: Mistakes in the Given Algorithm:

Wrong Algorithm (Given):

1. Define num1, num2, prod, and index.
 2. Enter num1 and num2.
 3. Set prod = 0 and index = 1.
 4. Repeat until index <= num2:
 - Add num1 to prod.
 - Increase index by 1.
 5. Print prod.
 6. Stop.
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- The algorithm only works when both numbers are positive.
 - It fails to handle numbers properly.
 - It does not check if either number is zero.
 - It always repeats the addition process num2 times even when fewer steps are enough.
 - It does not determine if the final result should be positive or negative.

Example:

Input: num1=3, num2=12

The algorithm adds 3 twelve times.

Output: 36

But it uses 12 additions instead of 3, so it is not efficient.

Task 2: Corrected Algorithm:

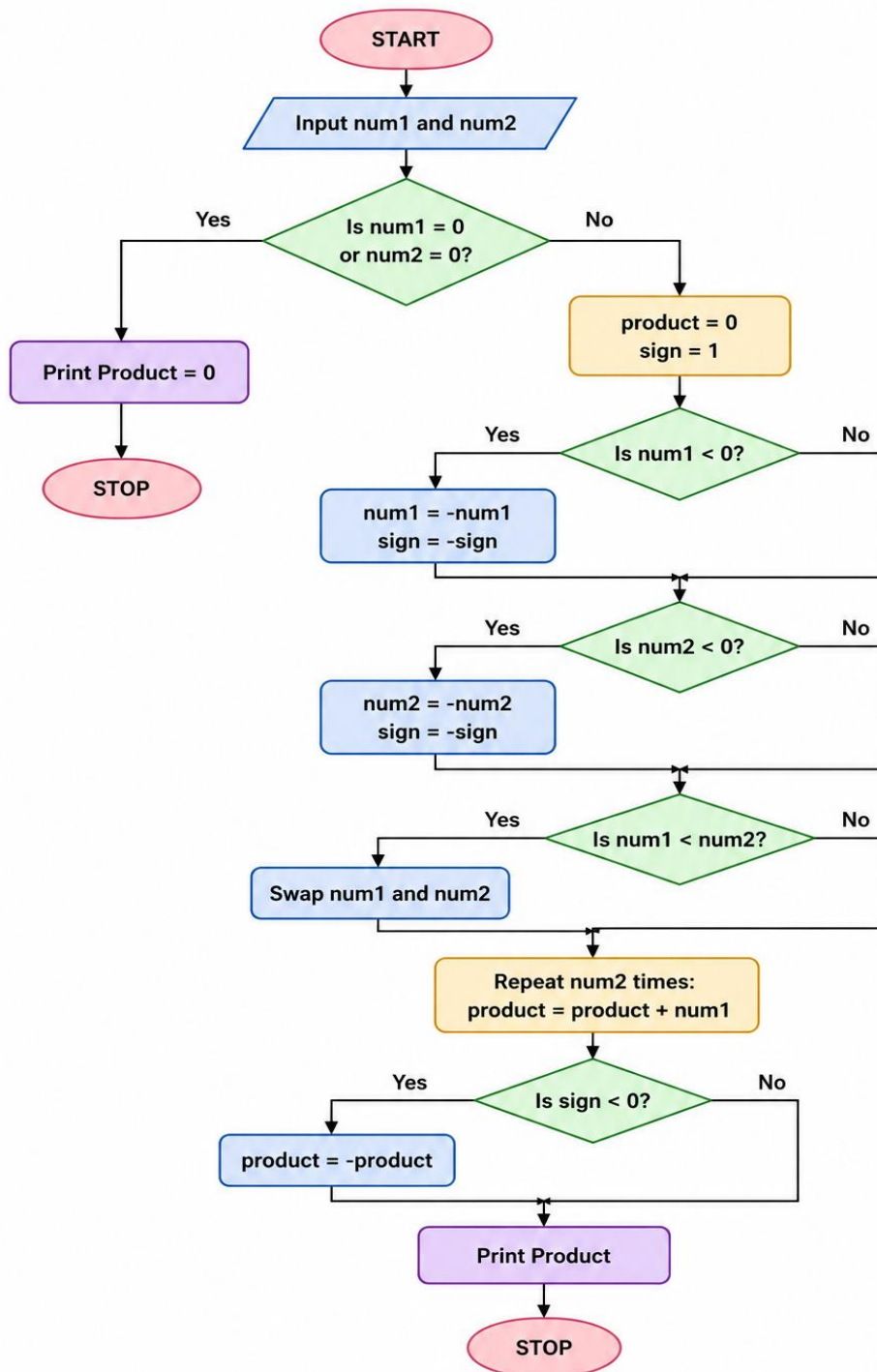
Algorithm to Multiply Two Numbers Without Using *

1. Start.
2. Declare num1, num2, product, and sign.
3. Enter the values of num1 and num2.
4. If either number is 0, print **Product = 0** and stop.
5. Set product = 0 and sign = 1.
6. If num1 is negative, make it positive and change the value of sign.
7. If num2 is negative, make it positive and change the value of sign.
8. Compare both numbers. If num1 is smaller than num2, swap them so that the loop runs fewer times.
9. Add num1 to product, num2 times.
10. If sign is negative, make the final product negative.
11. Display the product.
12. Stop.

Task 3: Mistakes in the Flowchart:

1. It does not check whether either number is zero.
2. It does not handle negative numbers.
3. It does not swap the numbers to reduce the number of additions.
4. It does not change the sign of the final answer when needed.

Task 4: Correct Flowchart :



Task 5: Verification:



Test Case	Input (num1, num2)	Expected Output	Minimum Additions	Result
1	12, 3	Product: 36	3	PASS
2	3, 12	Product: 36	3	PASS
3	-6, 5	Product: -30	5	PASS
4	6, -5	Product: -30	5	PASS
5	-9, -4	Product: 36	4	PASS
6	-4, -9	Product: 36	4	PASS
7	0, 8	Product: 0	0	PASS
8	8, 0	Product: 0	0	PASS
9	0, -7	Product: 0	0	PASS
10	-7, 0	Product: 0	0	PASS
11	0, 0	Product: 0	0	PASS
12	1, -7	Product: -7	1	PASS
13	-1, -9	Product: 9	1	PASS

Conclusion:

This assignment helped us understand how to multiply by adding without using the * symbol. We found mistakes in the algorithm and flowchart. Fixed them. We made sure the algorithm works with numbers, negative numbers and zero. We also improved the algorithm by using the possible additions, which makes it faster. We tested the corrected algorithm and flowchart with all test cases. Each test gave the expected result.